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Candidate surname	Other names
Master Functional Skills Model	Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	PMAT1/N01

Past Paper 2

Time: 25 minutes | Paper Reference **PMAT1/N02**

Mathematics

Level 1
Section A (Non-Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
14 /14

My signature confirms that I will not discuss the content of the test with anyone.

Signature: Master Functional Skills (MFS)

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided
  – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and your answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used.**

Information

- The total mark for this section is 14.
- The marks for each question are shown in brackets
  – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1 A charity wants to collect eighty four thousand pounds.

A charity worker says

> We have collected $\frac{3}{4}$ of the eighty four thousand pounds.

How much money has the charity collected?
You **must** show your working.

(3)

Step 1: Write eighty four thousand in digits:

Eighty four thousand = £84 000 [B1]

Step 2: Find 1/4 of £84 000 (divide by 4):

84 000 ÷ 4 = £21 000 [M1]

- 80 000 ÷ 4 = 20 000
- 4 000 ÷ 4 = 1 000
- 20 000 + 1 000 = 21 000

Step 3: Multiply by 3 to find 3/4 of the total:

21 000 × 3 = £63 000 [A1]

- 20 000 × 3 = 60 000
- 1 000 × 3 = 3 000
- 60 000 + 3 000 = £63 000

£ 63 000 [A1]

(Total for Question 1 is 3 marks)

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2 (a) Work out $28 - 20 \div 4$

(1)

Follow BIDMAS: Division before Subtraction:

- $20 \div 4 = 5$
- $28 - 5 = 23$ [B1]

23 [B1]

(b) Work out 6^2

(1)

Square number calculation (base × itself):

- $6^2 = 6 \times 6 = 36$ [B1]

36 [B1]

(c) Work out $-25 - 14$

(1)

Combining two negative values (add magnitudes, keep negative):

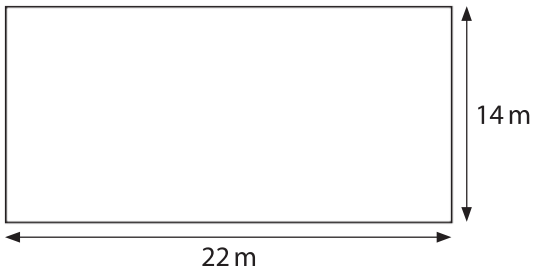
- $25 + 14 = 39$
- Therefore $-25 - 14 = -39$ [B1]

-39 [B1]

(Total for Question 2 is 3 marks)

3 Zeenat wants to buy edging to put along the edges of her lawn.
She has this sketch of the lawn.

The lawn is in the shape of a rectangle.



Each metre of edging costs £5
Zeenat thinks the total cost of the edging will be £360

(a) Is Zeenat correct?
Show why you think this.

(3)

Step 1: Calculate perimeter of lawn (22m by 14m):

Perimeter = 22 + 14 + 22 + 14 = 72 m [M1]

Step 2: Calculate total cost at £5 per metre:

Total cost = 72 × £5 = £360 [M1]

Step 3: Compare with Zeenat's estimate (£360):

Calculated cost (£360) = £360. Yes, she is correct!

Yes [A1]

(b) Use a reverse calculation to show a check of your answer.

(1)

Reverse calculation check on total cost:

£360 ÷ 5 = 72 m (matches perimeter) [B1]

Or: 360 ÷ 72 = £5 per metre

(Total for Question 3 is 4 marks)

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4 Gareth wants to know the amount of water needed to fill his pond.

The pond is in the shape of a cuboid as shown.



1 m^3 is 1000 litres.

Calculate the amount of water needed to fill the pond.
Give your answer in litres.

(4)

Step 1: Calculate volume of cuboid pond (length × width × depth):

Volume = 4 m × 2 m × 0.9 m [M1]

- Base area = 4 × 2 = 8 m^2
- Volume = 8 × 0.9 = 7.2 m^3 [M1]

Step 2: Convert cubic metres to litres (1 m^3 = 1 000 litres):

Litres = 7.2 × 1 000 = 7 200 litres [M1, A1]

Amount of water needed = 7 200 litres

7 200 [A1] litres

(Total for Question 4 is 4 marks)

TOTAL FOR SECTION A IS 14 MARKS

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Past Paper 2

Time: 1 hour 30 minutes | Paper Reference **PMAT1/C02**

Mathematics

Level 1
Section B (Calculator)



You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
42 /42

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- Check your working and your answers at each stage.
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- **Calculators may be used.**
- If your calculator does not have π button take the value of π to be 3.14

Information

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- The total mark for this paper is 56.
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Advice

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- Check your answers if you have time at the end.

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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 Here are 7 numbers.

3.51 3 1.1 4 4.5 2.5 2.81

Calculate the mean of these 7 numbers.

(3)

Step 1: Calculate the total sum of all 7 numbers:

3.51 + 3 + 1.1 + 4 + 4.5 + 2.5 + 2.81 [M1]

- 3.51 + 3.00 = 6.51
- 6.51 + 1.10 = 7.61
- 7.61 + 4.00 = 11.61
- 11.61 + 4.50 = 16.11
- 16.11 + 2.50 = 18.61
- 18.61 + 2.81 = 21.42

Total sum = 21.42

Step 2: Divide sum by 7 to find the mean:

Mean = 21.42 ÷ 7 [M1]

Mean = 3.06 [A1]

3.06 [A1]

(Total for Question 1 is 3 marks)

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2 In a factory the working week is from Monday to Friday.
There are 46 working weeks in a working year.

(a) Calculate the number of days in one working year.

(1)

Monday to Friday = 5 working days:

46 weeks × 5 days = 230 working days [B1]

230 [B1]

Nita works at the factory.
She will get a bonus if her absence rate is 2% or less.

This rule is used to calculate her absence rate.

number of days absent → ÷ number of days in the working year → × 100 → absence rate (%)

Nita was absent from work for 9 days in the last working year.

(b) Does Nita get a bonus?
Show why you think this.

(3)

Step 1: Calculate Nita's absence rate using formula:

Absence rate = (9 ÷ 230) × 100 = 3.913...% (≈ 3.91%) [M1]

Step 2: Compare with bonus threshold (≤ 2%):

3.91% > 2% (absence rate exceeds the 2% maximum limit).

Alternative check: 2% of 230 = 4.6 days. Since 9 > 4.6 days,

Nita was absent for too many days. She does NOT get a bonus.

No [A1]

(Total for Question 2 is 4 marks)

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3 Tina is getting broadband for her new flat.

She finds this deal.

Broadband deal

router normal price £39.99

15% discount on the normal router price

18 months contract at £56.99 each month

Tina buys the router and the 18 months contract.

Work out the total cost of this deal for Tina.

(5)

Step 1: Calculate 15% discount on the router (£39.99):

Discount = 15% of £39.99 [M1]

- 39.99 ÷ 100 × 15 = £5.9985
- Router cost = £39.99 − £5.9985 = £33.9915 [M1]

(Or: 39.99 × 0.85 = £33.9915)

Step 2: Calculate total monthly payments (18 months at £56.99):

Contract cost = 18 × £56.99 [M1]

- 18 × 56.99 = £1 025.82

(Working continued on next page...)

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Step 3: Calculate total cost of router + contract:

Total cost = Contract (£1025.82) + Router (£33.99) [M1]

- 1 025.82 + 33.9915 = £1 059.8115

Total cost = £1 059.81 (or £1 059.82) [A1]

Summary of costs:

- Discounted router: £33.99
- 18 months broadband: £1 025.82
- Overall package cost: £1 059.81

£ 1059.81 [A1]

(Total for Question 3 is 5 marks)

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4 Varney wants to cover a wall with a sealing mixture.
She reads this information about how to make the mixture.

Mix water with concentrate in the ratio 5 : 1

Varney wants to make 9 litres of the mixture.

How much water and how much concentrate does Varney need to make 9 litres of the mixture?

(3)

Step 1: Find total parts in ratio (water : concentrate = 5 : 1):

Total parts = 5 + 1 = 6 parts [M1]

Step 2: Find the volume of 1 part (total 9 litres):

1 part = 9 litres ÷ 6 = 1.5 litres [M1]

Step 3: Calculate volume of water and concentrate:

- Water (5 parts) = 5 × 1.5 = 7.5 litres [A1]
- Concentrate (1 part) = 1 × 1.5 = 1.5 litres [A1]

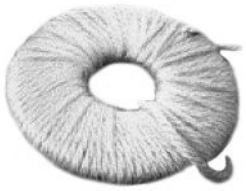
Check: 7.5 + 1.5 = 9.0 litres (matches total mixture volume):

water	7.5	[A1]	litres
concentrate	1.5	[A1]	litres

(Total for Question 4 is 3 marks)

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5 Joseph is making a template for craftwork.
He makes the template using two circles.



The two circles have the same centre.
The outer circle has radius 6 cm.

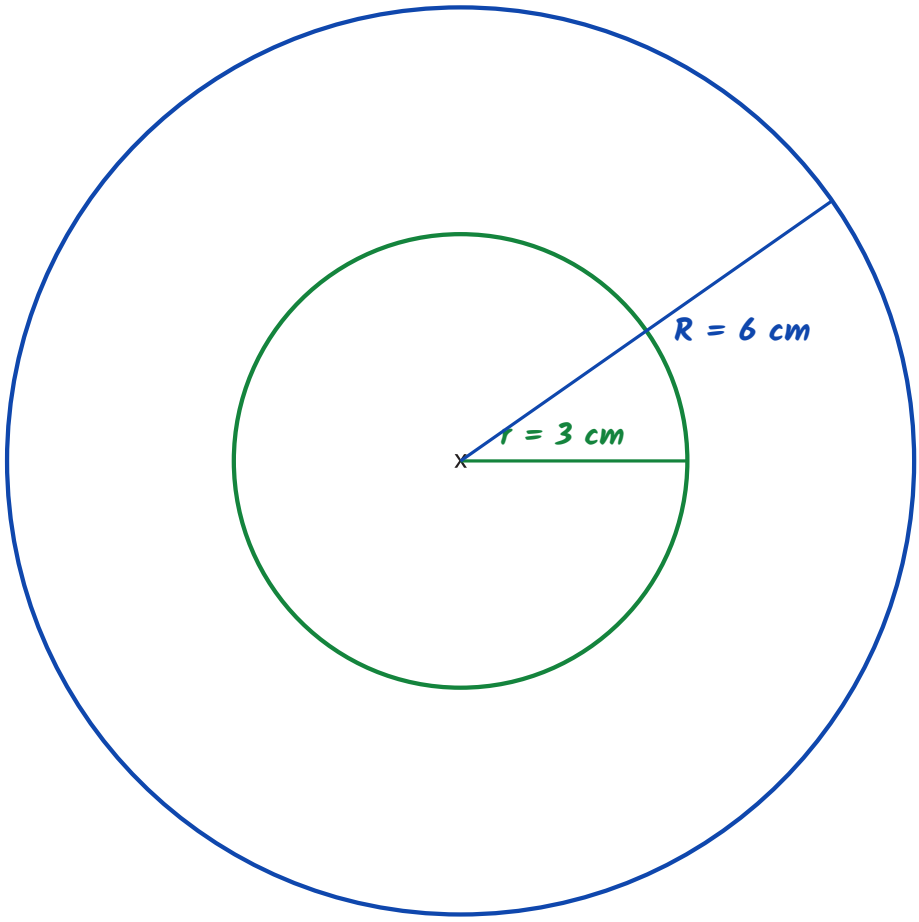
The radius of the inner circle is $\frac{1}{2}$ the radius of the outer circle.

Draw the template for Joseph in the space below.
The centre (x) is drawn for you.

(3)

Step 1: Calculate inner radius: 1/2 of 6 cm = 6 ÷ 2 = 3 cm [M1]

Step 2: With compass at centre (x), draw concentric circles: [M1, A1]



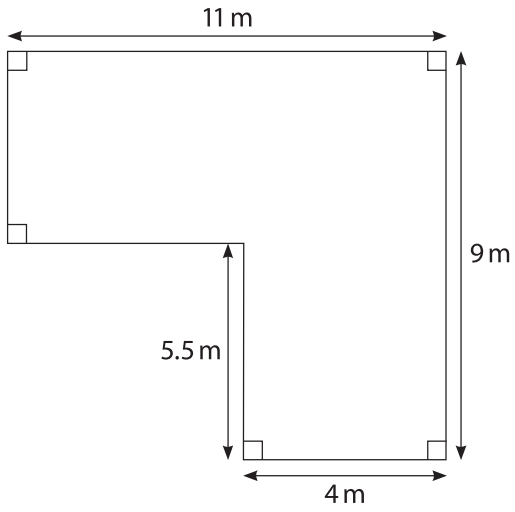
- *Inner circle: radius = 3 cm [M1]*
- *Outer circle: radius = 6 cm [A1]*

Template accurately drawn with compass at centre (x) [ABC]

(Total for Question 5 is 3 marks)

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6 Harpreet wants to put tiles on the ceiling of a room.
He has this sketch of the ceiling.



Harpreet knows one pack of ceiling tiles

- covers an area of 3.24 m^2
- costs £34.95

Ceiling tiles can be cut and joined.

Calculate the total cost of the packs of ceiling tiles Harpreet has to buy.

(6)

Step 1: Find missing lengths: 11 − 4 = 7 m; 9 − 5.5 = 3.5 m [M1]

Step 2: Split ceiling into two rectangles to find total area:

· Area A = 11 m × 3.5 m = 38.5 m^2; Area B = 4 m × 5.5 m = 22.0 m^2

Total area = 38.5 + 22.0 = 60.5 m^2 [M1]

(Check: 11 × 9 − 7 × 5.5 = 99 − 38.5 = 60.5 m^2)

(Working continued on next page...)

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Step 3: Number of ceiling tile packs needed (1 pack = 3.24 m²):

Number of packs = 60.5 ÷ 3.24 [M1]

60.5 ÷ 3.24 = 18.6728... packs

Since tiles come in full packs, round UP to 19 packs [M1]

Step 4: Calculate total cost of 19 packs at £34.95 each:

Total cost = 19 × £34.95 [M1]

- 20 × 34.95 = £699.00
- 1 × 34.95 = £34.95
- £699.00 - £34.95 = £664.05

Total cost = £664.05 [A1]

£ 664.05 [A1]

(Total for Question 6 is 6 marks)

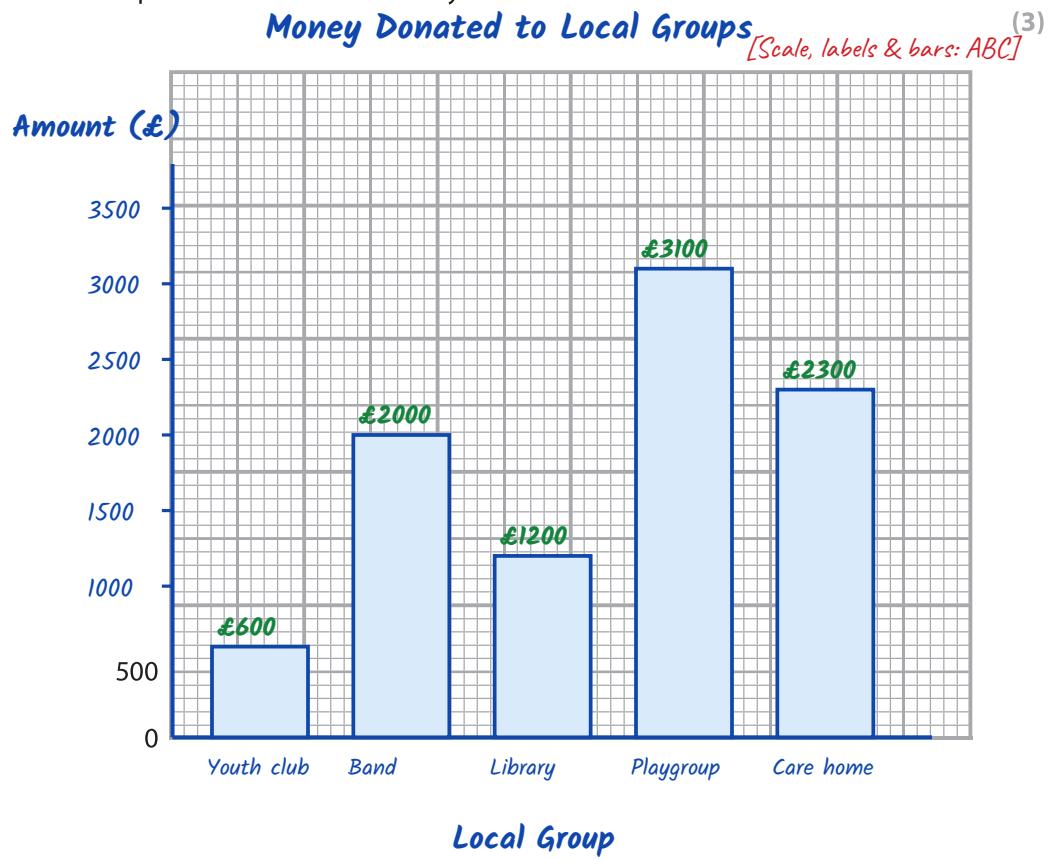
7 Grace works at a supermarket.
The supermarket gives money to local groups.

Grace wants to show customers the amount of money given to each local group.
She has this information.

local group	amount of money (£)
youth club	600
band	2000
library	1200
playgroup	3100
care home	2300

Draw a bar chart for this information.
Use 1 cm to represent £500 on the money axis.

(3)

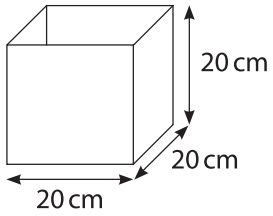


(Total for Question 7 is 3 marks)

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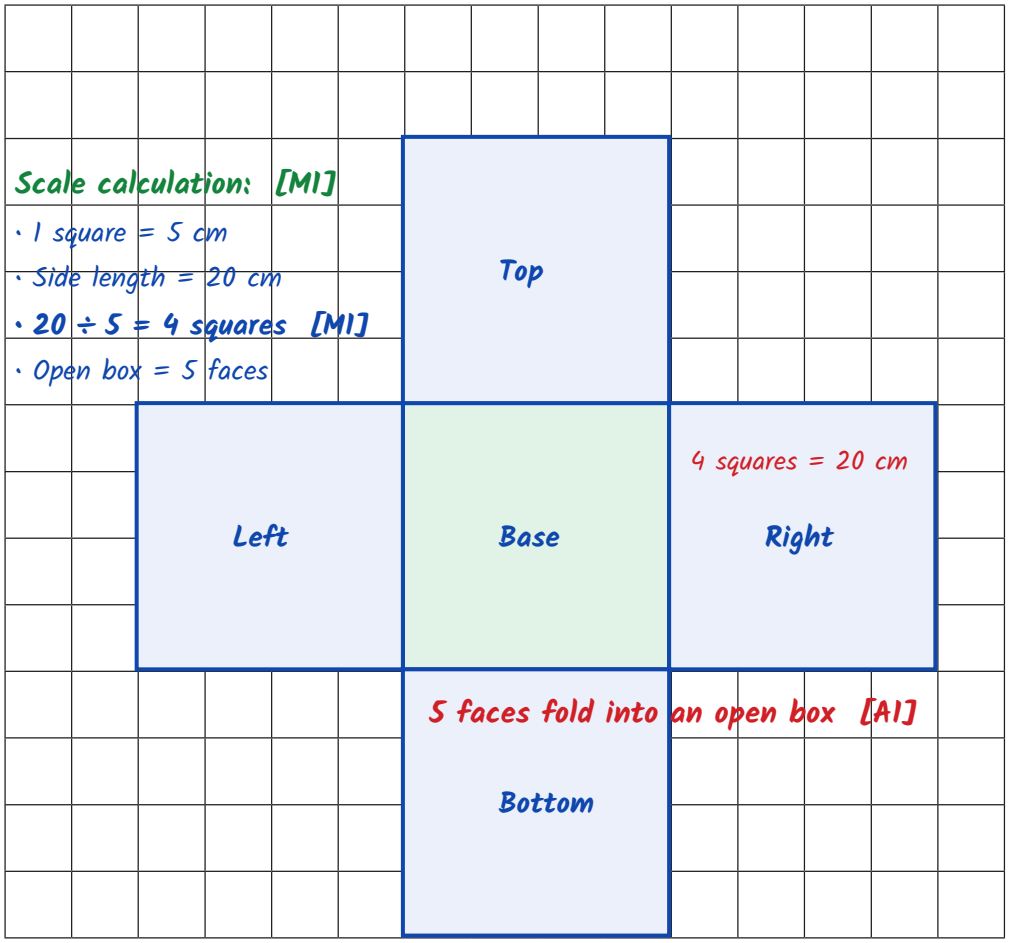
8 Jakob is a florist.
He designs an open box for flowers to stand in.

The box is in the shape of a cube with a side length 20 cm.



Draw a net of the open box.
Use the grid below to draw your net.

(3)



Key
1 square length on the grid represents 5 cm on the cube

(Total for Question 8 is 3 marks)

9 Sandro is making cottage pie.

He finds this list of ingredients.

cottage pie
serves 8 people
1500 grams of meat
250 ml stock
4 tbsp tomato paste
2 kilograms of potatoes

Sandro wants to make cottage pie for 12 people.
He wants to know how much meat to use.

(a) Calculate the amount of meat Sandro needs to use.

(3)

Step 1: Find amount of meat needed per person (8 people use 1 500 g):

1 500 g ÷ 8 = 187.5 g per person [M1]

Step 2: Multiply by 12 people:

187.5 g × 12 = 2 250 g [M1]

- *Alternative scaling factor: 12 ÷ 8 = 1.5*
- *1 500 g × 1.5 = 1 500 + 750 = 2 250 g [A1]*

Total meat required = 2 250 g

2 250 *[A1]* grams

☑ (b) Use a reverse calculation to show a check of your answer.

(1)

Reverse calculation check:

2 250 g ÷ 12 = 187.5 g per person [B1]

187.5 g × 8 = 1 500 g (matches original 8-person recipe)

(Total for Question 9 is 4 marks)

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10 Jenny is a science technician.
She carries out an experiment.

Jenny records the temperatures from her experiment each day.

day	1	2	3	4	5	6	7
temperature (°C)	65.5	65.2	66.5	69.0	70.0	71.8	68.2

(a) Calculate the range of these temperatures.

(2)

Identify highest and lowest recorded temperatures:

- *Highest temperature = 71.8 °C (Day 6)*
- *Lowest temperature = 65.2 °C (Day 2)*

Range = 71.8 − 65.2 = 6.6 °C [M1, A1]

6.6 *[A1]* °C

Jenny has included the range in her report.
She must write another comment about these temperatures in her report.

(b) Write a comment about the temperatures.

(1)

Tutor commentary on the experiment findings:

The temperature rose steadily from Day 2 (65.2 °C) to a peak on Day 6 (71.8 °C), then dropped on Day 7 (68.2 °C). [B1]

Alternative valid statistical comment (Mean):

- *Sum = 476.2 °C; Mean temperature = 476.2 ÷ 7 = 68.03 °C (≈ 68.0 °C)*

(Total for Question 10 is 3 marks)

11 Clare sells houses.
She has this information about the purchase prices of some houses.

house reference	purchase price (£)
A	123 950
B	249 000
C	94 500
D	389 000
E	121 000
F	149 000

(a) Write in order the purchase prices of these houses.
Start with the lowest price.

(1)

Ascending order (from lowest to highest price):

1. £94 500 (House C - 5 digits)
2. £121 000 (House E)
3. £123 950 (House A)
4. £149 000 (House F)
5. £249 000 (House B)
6. £389 000 (House D)

Ordered: 94 500, 121 000, 123 950, 149 000, 249 000, 389 000 [B1]

People pay stamp duty on houses with a purchase price of more than £125 000

(b) Tick the purchase prices that are greater than £125 000

(1)

purchase price (£)	
123 950	
249 000	✓
94 500	
389 000	✓
121 000	
149 000	✓

> 125k

> 125k

> 125k [B1]

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Alex buys a house.
He must pay £255 000 for the house.
He must also pay 5% of the purchase price in stamp duty.

(c) Calculate the total amount Alex must pay.

(3)

Step 1: Calculate 5% stamp duty on £255 000:

Stamp duty = 5% of £255 000 [M1]

- 255 000 ÷ 100 = £2 550 (1%)
- £2 550 × 5 = £12 750 (5%)

Stamp duty = £12 750

Step 2: Calculate total amount to pay (Price + Stamp duty):

Total amount = £255 000 + £12 750 [M1]

Total amount = £267 750 [A1]

Alternative single-step percentage multiplier method:

100% + 5% = 105% = 1.05

£255 000 × 1.05 = £267 750

£ 267 750 [A1]

(Total for Question 11 is 5 marks)

TOTAL FOR SECTION B IS 42 MARKS
TOTAL FOR PAPER IS 56 MARKS

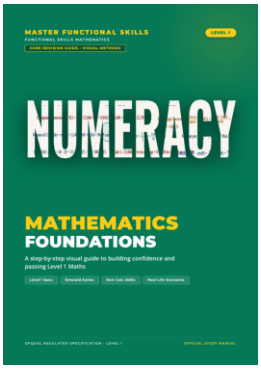
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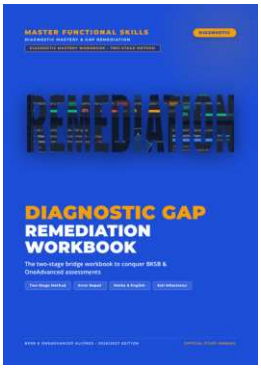


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